ATHARVA GITE

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EDUCATION

King's College London - London, England — Sep 2026 - Expected Aug 2027
MSc, Advanced Computing

Sardar Patel Institute of Technology - Mumbai, India — 2021 - 2025
Bachelor of Technology, Computer Engineering — *CGPA: 8.48/10*
Relevant Coursework: Data Structures, Algorithms, Operating Systems, Distributed Computing, Computer Networks, DBMS, Software Engineering, AI/ML, Cloud Computing, Systems Programming

EXPERIENCE

Founding AI Engineer, *Celeris* — Oct 2025 - Sep 2026

- Led migration of an OCR pipeline for handwritten/scanned medical and pharma documents from local Pytesseract/PyMuPDF processing to Mistral's OCR-4 API, cutting processing time from 40s to 12s and lifting extraction accuracy from 0.47 to 0.83 for files under 10MB.
- Cut caller-perceived response latency by about 70% on VocaLink by streaming synthesized speech as it was generated and stopping the agent mid-sentence when the caller interrupted.
- Integrated AI-assisted contract generation into a Contract Lifecycle and Finance Management platform (Python, Express, MongoDB, React), building reusable APIs across contract, finance, and asset modules.
- Unified phone, SMS, and web chat through one request engine that completes bookings, leads, and tickets within the same conversation, with request decisions at 60–125ms against a 700ms turn budget.

Python Development Intern, *IDMS InfoTech* — Jan 2025 - Jun 2025

- Reduced automated Minutes of Meeting generation time from approximately 3 minutes to 40 seconds by optimizing production transcription and processing workflows.
- Implemented multilingual meeting processing supporting 70+ languages and automated the workflow from transcription and translation through meeting summarization and email distribution.

Teaching Assistant, *SPIT* — Aug 2024 - May 2025

- Guided 200+ undergraduate students in Ubuntu/Linux, terminal workflows, debugging, operating systems, SQL/MySQL, and DBMS concepts during laboratory sessions.

PROJECTS

Limit Order Book Simulator — *C++20, CMake, Catch2, Google Benchmark*

- Built a deterministic C++20 limit-order-book and matching engine with price-time FIFO, market sweeps, iceberg replenishment, and pro-rata allocation, modeling exchange-style queue priority and fills.
- Designed a single-writer architecture with bounded MPSC ingress and a preallocated slab order pool; order adds were ∼2.4× faster than heap allocation, validated with ThreadSanitizer and benchmarks.

Folio Job Application Tracker — *TypeScript, Next.js, React, PostgreSQL, Vitest, Playwright*

- Engineered a multi-user application platform with PostgreSQL-backed state/history, server-side ownership enforcement, CSV import, private resume storage, and dashboard analytics using shared service/repository logic.
- Made application state changes transactional by recording status, history, and audit events together; built unit/integration tests, Playwright workflows, and CI covering type checks, tests, and production builds.

StudyForge Course RAG System — *Python, FastAPI, PostgreSQL, pgvector, Next.js, OpenAI*

- Built a course-scoped study system that extracts and chunks uploaded files, embeds them, and generates grounded Q&A, flashcards, quizzes, and exam plans from active document versions only.
- Fused pgvector similarity with PostgreSQL full-text search, withheld generation when retrieval was too weak, and implemented leased ingest jobs with retry and fencing for concurrent workers.

Yield Curve Engine — *Python, NumPy, SciPy, pandas, scikit-learn, Docker, pytest*

- Built a fixed-income research engine comparing Nelson-Siegel, Nelson-Siegel-Svensson, and smoothing spline curve fits, including nonlinear decay optimization and leave-one-maturity-out validation.
- Applied PCA to daily yield changes across 11 Treasury maturities and implemented walk-forward 10s2s and 2s5s10s spread backtests with lagged signals, while maintaining reproducibility through Docker and 18 tests.

Pairs Trading Statistical Arbitrage Engine — *Python, pandas, NumPy, scikit-learn, pytest*

- Built a statistical-arbitrage research pipeline comparing static OLS, causal rolling OLS, and Kalman hedge ratios with lagged exposures for walk-forward backtesting.
- Applied Benjamini-Hochberg FDR/Bonferroni corrections, half-life analysis, and fixed discovery/live splits; modeled dollar positions with turnover, transaction and borrow costs, net market beta, and trade-level blotters.

SKILLS

Languages: Python, C++, JavaScript/TypeScript, Java, SQL, R
Frameworks: Node.js, Express.js, React, Next.js, FastAPI, Flask
Databases: PostgreSQL, MongoDB, MySQL, pgvector
AI & Data: OpenAI, LLM Integration, RAG, Apache Spark, Hadoop, Hive, pandas, NumPy, scikit-learn
Software Engineering: Backend Development, REST APIs, OOP, Testing, CI/CD, Debugging, SDLC, Git
Systems & Cloud: Linux, Bash, Docker, AWS, GitHub Actions, Distributed Systems, Networking, Concurrency
Developer Tools: Postman, pytest, Vitest, Playwright, CMake, Catch2, Google Benchmark

CERTIFICATIONS

- **Mathematics for Machine Learning Specialization Imperial College London**
- **Sports Performance Analytics Specialization University of Michigan**
- **Google Data Analytics Google**
- **SPIT Barclays Cyber Shala** 2023-2024

POSITIONS OF RESPONSIBILITY

- **Head of Events, OCULUS IPL Auction** Aug 2023- March 2024
  Led a team of 20+ members to organize a flagship IPL auction event for 500+ participants, coordinating logistics, pricing, and execution workflows. Managed event operations and coordination to ensure smooth execution and participant engagement.
- **Head of Operations, OCULUS IPL Auction** Aug 2022-March 2023
  Coordinated logistics, participant flow, and operational execution to ensure smooth event management and timely setup. Managed resource planning and cross-team coordination to improve operational efficiency during high-participation activities.
- **Head Of Operations, OCULUS Aelaan-e-Jung** Aug 2022- March 2023
  Managed inventory, logistics, and participant coordination to ensure timely availability of event resources and smooth execution. Coordinated vendor and operations workflows to meet financial and operational timelines effectively.

COMMUNITY WORK

- Conducted sessions on basic computer literacy for primary school students at a local government school. 2025
- Fundraising volunteer for Child Rights & You (C.R.Y.) under the Oculus for a Cause 2023 campaign 2023
- Volunteered at the Open Mic event organized by SPARK in 2023 and 2024, assisting with event coordination and ensuring a smooth experience for participants and attendees.
- Volunteered at eSummit, organized by E-Cell SPIT, in 2023 and 2024, assisting with event coordination and supporting various activities to ensure the success of this entrepreneurial summit.

EXTRA CURRICULARS & SPORTS ACHIEVEMENTS

- **Winners,General Championship 2023**: As Vice-Captain of the Computer Engineering (CE) department, guided and inspired the sports, tech, and cultural teams, whose collective efforts were instrumental in the department securing the "Best Performing Department" award for excellence in both academic and extracurricular pursuits.